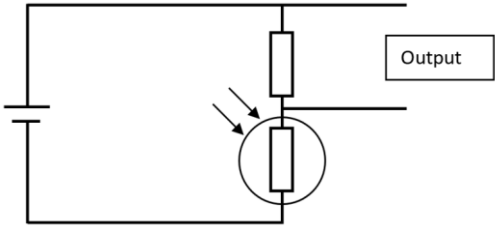


		(ii)	$9 = I(1200)$ $I = 7.5 \times 10^{-3} \text{ A}$ $I = nAve$ $V = I/(nAe) = 7.5 \times 10^{-3} / (2.2 \times 10^{28} \times 1 \times 10^{-3} \times 3.13 \times 10^{-5} \times 1.6 \times 10^{-19})$ $= 6.81 \times 10^{-5} \text{ m s}^{-1}$	
	(b)	(i)	Light Dependent Resistor ("LDR" not accepted)	

		(ii)		
		(iii)	The resistor and the light-dependent resistor are connected in series with the cell so that the cell's e.m.f. is shared between them. Thus, when the potential difference of one of the component increases, the other should decrease using the potential divider principle. The light dependent resistor characteristic is such that when the ambience light is bright, the resistance is low, thus the potential difference across it will be low and the potential difference across the resistor will be high, and vice versa.	
	(c)	(i)	E.m.f. of a source is defined as the amount of energy converted from other forms of energy to electrical energy when the source drives a unit charge round a complete circuit. Whereas for potential difference, it is energy converted from electrical to other forms of energy.	
		(ii)	Since there is not current in the ammeter, the potential difference across XM and YN must be the same, and the potential at X must be the same as Y.	
		(iii)	$V_{10} = 12 \times 10/25 = 4.8 \text{ V}$ Thus $V_L = 4.8 = 5 \times L/25$ $L = 24 \text{ cm}$	
		(iv)	If the 12 V cell has internal resistance, the EMF of the cell will have to drop some p.d. across the internal resistance. Thus the p.d. across V_{10} will decrease. For the ammeter to continue to have a null deflection, length L will need to be decreased. OR For the ammeter to continue to have a null deflection, thus the length between YP must increase	